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Evaluating fabrication feasibility and biomedical application potential of in situ 3D printing technology

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1. Introduction

Tissue engineering is an interdisciplinary realm that utilizes engineering and life science principles to advance the knowledge of tissue growth and behavior. This knowledge can be applied further to develop biological substitutes to restore, maintain, or improve tissue function. [1] The architecture of native tissues in three-dimensional (3D) is highly hierarchical with heterogeneous cells and bioactive factors distributed in patterns. This architecture was found to be critical to tissue growth and function. Thus, it is a major challenge in tissue engineering to fabricate structures which mimics this 3D architecture.

Charles Vancanti *et al.* demonstrated a therapeutic technique in which a tissue-engineered cartilage scaffold was engineered into the shape of a human ear. [2] Similar cartilage structures were then engineered as: temporomandibular disc [3] and joint [4], meniscus [5], trachea [6], intervertebral disc [7], nasal tip [8], and nasal septum [9], as well as other bone structures as: femoral shaft [10], mandibular condyle [11] and distal phalanx [12]. Though anatomical geometries were created, these fabricated tissue scaffolds lack the ability to encapsulate cells within the structure. To overcome this, specific fabrication methods and techniques were developed to utilize cells to fabricate 3D laden construct with bioactive components. These includes, but is not limited to: inkjet [12], [13], laser-guided printing [14], two-photon polymerization and laser induced forward transfer [15], and 3D bioactive Solid Free-form Fabrication (SFF) [16]–[20].

SFF is the technology analogous to 3D printing. 3D bioactive SFF technologies have the unique capabilities to fabricate complex geometries with living cells while maintaining high cell viability. By encapsulating various types of live cells and bioactive component within 3D printed construct, it enables the spacious allocation of the cells to better mimic the anatomy of human tissues. Cohen *et al.* developed an approach to fabricate cell-laden constructs using SFF technologies by mixing alginate with calcium sulfate as the deposition material. [16] This pre-cross-linked hydrogel was then mixed with cells and extruded to produce his cell-laden tissue constructs. Ahn *et al.* modified Cohen's approach by introducing an aerosol cross-linking process to add calcium chloride (CaCl_2) to the cell-alginate suspension. [17] During the printing process, Ahn fumed CaCl_2 with an ultrasonic humidifier to solidify the tissue construct. Khalil *et al.* utilized an alternative method [18] which involves mixing alginate with cells followed by extrusion to a reservoir filled with CaCl_2 . Sequentially, CaCl_2 was added into reservoir as layers build up. In addition to ionic cross-linking, a temperature control system was used for the solidification process. Ahn *et al.* developed a direct cell printing approach with a low-temperature processing unit. [19] During the fabrication process, the alginate based material was extruded at a cooled working stage for structural fabrication. The fabricated scaffold was subsequently cross-linked with CaCl_2 . Zhao *et al.* utilized a temperature controlled system for 3D cellular laden construct fabrication. [20] Zhao used gelatin/alginate/fibrinogen hydrogel that were pre-mixed with cells as the cell delivery material. The cell suspension was loaded into a temperature controlled deposition system at 25 °C before extrusion and at 10 °C after extrusion. This temperature difference is essential for the solidification process of gelatin, hence producing a porous tissue scaffold with living cells. Those mentioned fabrication approaches are sensitive to solidification time, temperature of controlled environment, or printing stage in the case of reservoir cross-linking process.

This article presents an *in situ* fabrication method which enables simultaneous cross-linking at ambient environmental condition. This method does not require a temperature control unit or a reservoir printing

stage. A dual-nozzle component was developed to simultaneously print and cross-link alginate suspension to fabricate desired tissue 3D scaffolds. Alginate suspension and cross-linking solution were held in the dual nozzles. Struts were fabricated by the simultaneous extrusion from both nozzles resulting in cross-linked alginate hydrogel. The fabricated struts were printed into patterned layers. 3D scaffolds were fabricated as layers built up. Fabrication parameters were investigated in regarding to strut formability (the capability to fabricate a cylindrical strut in the same diameter as dispensing tip), structural stability (the capability to hold its structure against mechanical disturbance), and structural integrity (the capability to preserve its structure in cell culture condition). The structural stability and integrity were examined to find fabrication parameters that enable a stable structure for future biomedical application and experimental handling. To evaluate strut formability, both deposition related parameters (alginate flow rate, dispensing tip size, and nozzle travel speed) and cross-linking related parameters (alginate concentration, calcium concentration and cross-linker to deposition flow rate ratio) were investigated. 3D scaffolds were fabricated using parameters that achieved the strut formability and were investigated for structural stability and integrity. Strut formability was found can be achieved when the flow rate of alginate suspension and nozzle travel speed are set according to the dispensing tip size in the condition of sufficient cross-link. Cross-linking sufficiency range was determined for cross-linking parameters. The structural stability and structural integrity were controlled by alginate composition. A range of the fabrication parameters were determined for fabrication of a desired stable scaffold with structural integrity for 14 days. These findings are presented to demonstrate the potential in biomedical applications.

2. Methods and materials

2.1. Dispensing system

A 3D bio-printer system was used to fabricate 3D scaffolds [18], [21]. An image of the system is shown in Figure 1. The 3D bio-printer system operates at room temperature and low-pressure conditions. This system includes a 3D spatial control component and a multi-nozzle deposition component. The 3D spatial control component is composed of three orthogonal arms (Parker Hannifin Corporation Automation Group, Cleveland, OH) that move accordingly to pre-defined 3D pathway. The path can be programmed using an in-house software that is designed for CAD models. The spatial control component resolution is 10 μm in the three orthogonal directions (x, y, z) and has minimum velocity of 100 $\mu\text{m/s}$.

Pneumatic micro-valves (EFD, East Providence, RI) are used for multi-nozzle deposition component. The nozzles are mounted on the z motion arm with connectivity to the x and y arm for 3D motion. Dual-nozzle extrusion was utilized for this study. Nozzle A extrudes the alginate suspension while Nozzle B extrudes the cross-linking solution. The dual-nozzle extrusion tips were brought together for *in situ* cross-linking solidification. 3D scaffold was fabricated along the motion path.

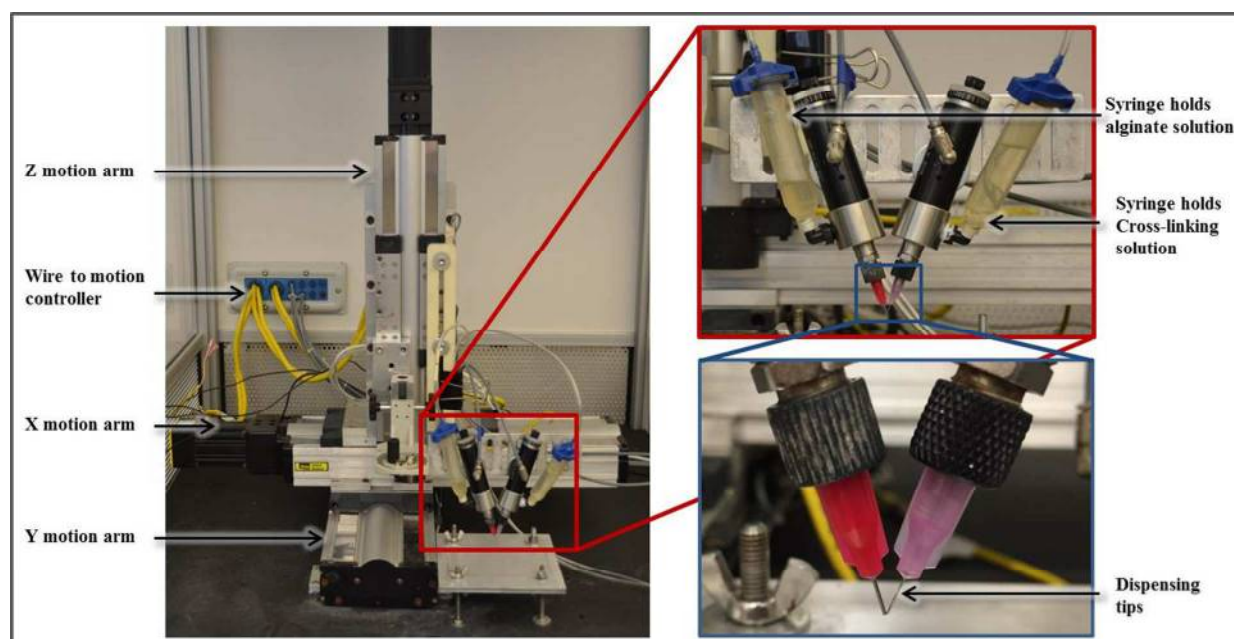


Figure 1. (Left) An image of the 3D bio-printer system, (right top) a close-up of the multi-nozzle deposition component, and (right bottom) a close-up of the dispensing tips

2.2. Fabrication protocol

Alginate A (Mw: 28 g/mol) (Sigma, St. Louis, MO) was mixed with alginate B (Mw: 100,000-200,000 g/mol) (Sigma, St. Louis, MO) to create the alginate suspension. The mixing ratio was examined in structural stability and integrity investigation. The mixed alginate was then weighted and dissolved in 0.1 M HEPES (4-(2-Hydroxyethyl) piperazin-1-ylethanesulphonic acid) (Electron Microscopy Science, Hartfield, PA). The alginate suspension was stirred overnight to ensure homogeneity in concentration. The alginate concentration was examined in cross-linking parameter investigation. The alginate suspension was then transferred to Nozzle A with a dispensing tip (EFD, East Providence, RI) assembled. The sizes of dispensing tips were examined in deposition parameter investigation. To create the cross-linking solution, ACS grade anhydrous calcium chloride (CaCl_2) (Amresco, Solon, OH) was weighted and dissolved in 0.1 M HEPES. The CaCl_2 concentrations were examined in cross-linking parameter investigation. The CaCl_2 suspension was then loaded on Nozzle B with tip size of 100 μm . Nozzles A and B were mounted onto the z axis of motion arm. The two nozzles were angled so that the tips were positioned close to each other, as shown in the close-up of Figure 1. Both nozzles were angled symmetrically to form a 30 degrees angle between them. This positional setup would allow the tips outlets facing to each other for mixing extrusions, and facing downwards for lying down strut. When the alginate and the cross-linking solution were extruded, the close proximity of the tips allowed the alginate and the CaCl_2 solution to mix. As a result, hydrogel was created on the printing substrate. Pneumatic pressure was tuned accordingly to control suspension extrusion flow rate. Alginate suspension flow rates were examined in deposition parameter investigation. CaCl_2 suspension flow rate were controlled as a ratio to alginate suspension flow rate. CaCl_2 to alginate flow rate ratios were examined in cross-linking parameter investigation. Alginate hydrogel strut was fabricated as continuous extrusion of hydrogel as nozzles traveled. Travel speeds were examined in deposition parameter investigation. 3D scaffolds were fabricated as layer- by- layer fashion. Hydrogel struts were patterned in parallel for each layer. The

pattern was designed with an outer dimension of 15 mm × 15 mm with a 1 mm space between each strut. Pattern direction was changed orthogonally between adjacent layers. 10 layers were stacked to form the 3D scaffold with overall height of 1.5 mm. (Figure 2.)

Since seven fabrication parameters were investigated in this study, it was impractical and unnecessary to examine all the fabrication parametrical value combinations. The investigation methodology was described as follows: A set of base fabrication parameter values were utilized as control settings. During each of the parametric investigation, a set of examination values of the parameter was studied while other parameters were kept as base values. (Table 1.) Along with the series of investigations, examination values that fail to achieve the investigation criteria would be eliminated for further investigation. Fabrication parametrical values were narrowed down for desired filament formation, structural stability and integrity. The base deposition related parameters values were partially determined by previous research of the effect of pneumatic pressure and dispensing size on cell viability. [21] Dispensing tip size of 250 μm was shown to result decent cell viability and recovery rate and thus was recruited as the base value in this study. The base crosslinking related parameters values were partially based on previous research of bio-printing 3D alginate cell laden construct using method where cell alginate mixture was deposited to CaCl_2 filled reservoir. [18]

Fabrication Parameters	Base Values	Examination Values
Alginate Flow Rate (mm^3/min)	23.5	22, 23.5 and 25
Dispensing Tip Size (μm)	250	200, 250 and 330
Nozzle Travel Speed (mm/s)	8	4, 8 and 12
Alginate Concentration (w/v)	4.5%	3.0%, 4.5% and 6.0%
CaCl_2 Concentration (w/v)	3.0% *	1.0%, 3.0% and 5.0%
CaCl_2 to Alginate Flow Rate Ratio (v/v)	3:1 *	1:1, 2:1, 3:1, 4:1 and 5:1
Alginate Mixing Ratio (Alginate A: B) (w/w) †	3:1	A only, 3:1, 2:1, 1:1 and B only

Table 1. Fabrication parameters were investigated

*: In deposition parameter investigation, CaCl_2 suspension concentration was set to 5%, and CaCl_2 to alginate flow rate ratio was set to be 5: 1 (v/v) to ensure a sufficient cross-linking condition,

†: Alginate mixing ratio was examined in structural stability and integrity investigation.

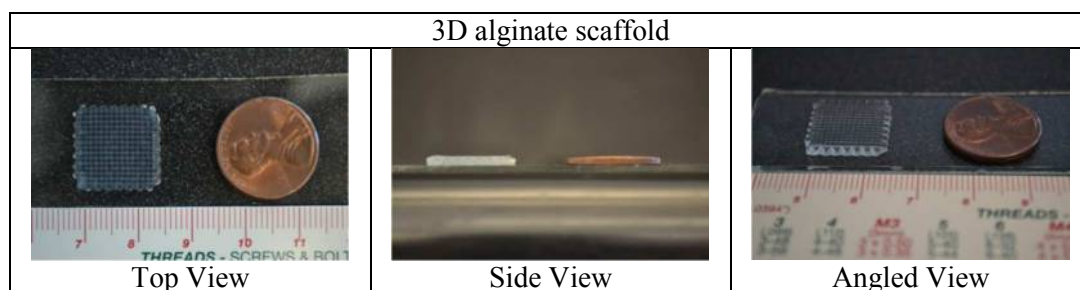


Figure 2. Top view, side view and angled view of 3D printed scaffold

2.3. Deposition parameter investigation

In order to reach strut formability (the capability to fabricate a cylindrical strut in the same diameter as dispensing tip) when cross-linked sufficiently, the flow rate of alginate suspension and nozzle travel

speed need to be set according to the dispensing tip's diameter. Equation (1) was developed by Khalil *et al* to describe this relationship. [18] Deposition parameters were investigated using Equation (1).

$$D = \sqrt{\frac{4Q}{v \cdot \pi}} \quad (1)$$

where D is the dispensing tip diameter, Q is the volume flow rate of alginate suspension, and v is the nozzle travel speed. The volume flow rate Q was examined in this study. It can be controlled by tuning the pneumatic pressure applied to nozzle according to Poiseuille's law as described in Equation (2).[22]

$$Q = \frac{\pi \cdot P \cdot R^4}{8 \cdot \eta \cdot L} \quad (2)$$

where Q is the volume flow rate, P is the applied pressure at the tip's inlet (inner orifice of the dispensing tip), R is the radius of dispensing tip, η is the viscosity of the material and L is the length of the tip. The relationship between pneumatic pressure applied and the pressure at the tip's inlet was described as Equation (3) [23].

$$P_1 = \frac{P_2 \cdot r^2}{R^2} \quad (3)$$

where P_1 is the pneumatic pressure applied to the syringe, P_2 is the pressure at the tip's inlet, r is radius of the inner orifice of the dispensing tip and R is the radius of the nozzle syringe.

Investigation of deposition parameter was performed as following: struts were fabricated using base fabrication parameter values where D, Q and v were chosen to satisfied equation (1). The diameter of fabricated strut was measured using a Leica optic microscope (Leica Microsystem, Bannockburn, IL) under 10 $\times$ magnifications and compared with the desired diameter of 250 μ m. Subsequently, struts were fabricated by individually altering D, Q and v to values below and above the base value while keeping the other parameters as base. The values below and above the base values were 200 and 330 μ m for D; 22.0 and 25.0 mm³/ min for Q; and 4 and 12 mm/s for v.

Sample size of five (n=5) was examined for each parametric setting in the deposition parameter investigation.

2.4. Cross-linking parameter investigation

The cross-linking related parameters: alginate suspension concentration, CaCl₂ suspension concentration, and CaCl₂ to alginate suspension flow rate ratio, were investigated. In cross-linking sufficient condition, strut formability was achieved when equation (1) was satisfied. Thus, strut formability was used as the criterion for cross-linking sufficiency examination.

Alginate suspension concentration was examined as 3.0%, 4.5% and 6.0% (w/v). CaCl₂ suspension concentration was examined as 1.0%, 3.0% and 5.0% (w/v). CaCl₂ to alginate suspension flow rate ratio was examined as 1:1, 2:1, 3:1, 4:1 and 5:1. Alginate suspension concentration was examined when the other parameters were kept as base. Since CaCl₂ suspension concentration and CaCl₂ to alginate suspension flow rate ratio were correlated to each other. Each pair combination between CaCl₂

concentration and CaCl_2 to alginate flow rate ratio were examined when other parameters were kept as base. Alginate hydrogel struts were fabricated and the diameters were measured using Leica optic microscope under $10\times$ magnifications. The measurements were then compared with desired strut size: $250\text{ }\mu\text{m}$.

Sample size of five ($n=5$) was examined for each parametric setting in the cross-linking parameter investigation.

2.5. Structural stability and integrity investigation

The material composition and fabrication parameters that achieve strut formability were carried on for structural stability and integrity investigation. The material composition was examined as alginate A only, alginate B only, and alginate A to alginate B ratio as 1:1, 2:1, and 3:1 (w/w). The cross-linking parameters were examined as following: alginate suspension concentration: 4.5% (base value) and 6.0%; CaCl_2 suspension concentration: 3% (base value) and 5%; CaCl_2 to alginate suspension flow rate ratio: 3:1 (base value) and 5:1. 3D scaffold were fabricated by altering each of the examining parameter when the other parameters were kept as base.

Stability investigation was designed to examine the capability of the 3D scaffold to hold its structure against gentle mechanical disturbance. It was examined as following: the fabricated 3D scaffolds were fabricated and moved to 35 mm petri dish with Dulbecco's phosphate- buffered saline (DPBS) (Mediatech, Manassas, VA) immersed. Gentle mechanical disturbance was introduced by placing the petri dish on an orbital shaker with 100 rpm for 5 min. The structures before and after disturbance were compared qualitatively.

3D scaffolds that sustained their structures against mechanical disturbance were examined for integrity in cell culture condition. Integrity investigation was designed to examine the capability of the 3D scaffold to preserve its integrity in cell culture conditions. It was examined as following: 3D scaffolds were immersed in cell culture medium and incubated in $37\text{ }^\circ\text{C}$ with 5% CO_2 . Cell culture medium was composed of 89% Dulbecco's modified eagle medium, 10% fetal bovine serum, 1% penicillin-streptomycin solution (Mediatech, Manassas, VA). The structural integrity was examined at day 0, 1, 3, 5, 7 and 14 qualitatively. The fabrication parameters pass the qualitative examination were further confirmed with a quantitative examination. It was conducted by measuring the struts diameters using Leica optical microscope under $10\times$ magnifications at day 0, 1, 3, 5, 7 and 14. Struts were measured at three random locations of each experiment sample.

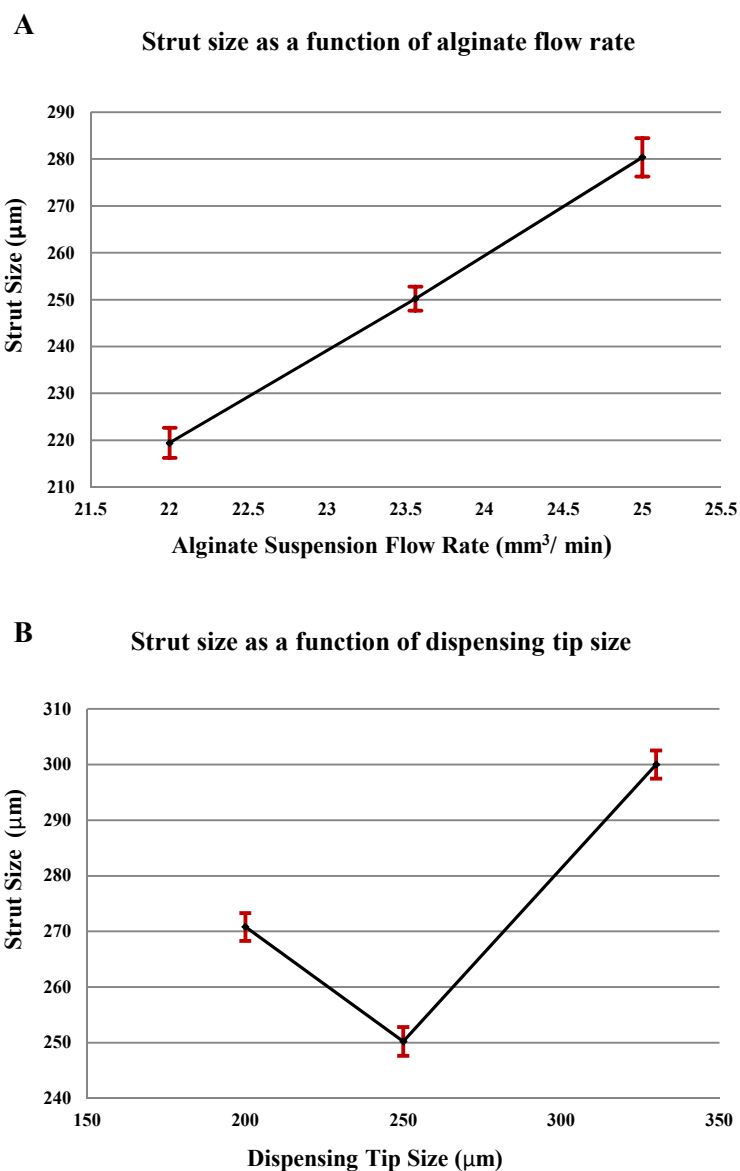
Sample size of three ($n=3$) was examined for each parametric setting in the structural stability and integrity investigation.

3. Result

3.1. Deposition parameter investigation

Effect of deposition parameters (alginate flow rate, dispensing tip size and nozzle travel speed) on strut formability was investigated. The fabricated struts were examined using Leica optical microscope with $10\times$ magnifications. The results were shown in Figure 3.

When the three parameters satisfied the equation (1): $Q = 23.5 \text{ mm}^3/\text{min}$, $D = 250 \text{ }\mu\text{m}$ and $v = 8 \text{ mm/s}$, the struts sizes were $250.20 \pm 2.59 \text{ }\mu\text{m}$. When altering Q to lower and higher flow rates, the struts sizes were $219.40 \pm 3.21 \text{ }\mu\text{m}$ and $280.40 \pm 4.10 \text{ }\mu\text{m}$ respectively. When altering D to smaller and larger diameters, the struts sizes were $270.80 \pm 2.49 \text{ }\mu\text{m}$ and $300.00 \pm 2.55 \text{ }\mu\text{m}$ respectively. When altering v to lower and higher speeds, the struts sizes were $372.80 \pm 7.05 \text{ }\mu\text{m}$ and $203.20 \pm 3.56 \text{ }\mu\text{m}$ respectively.



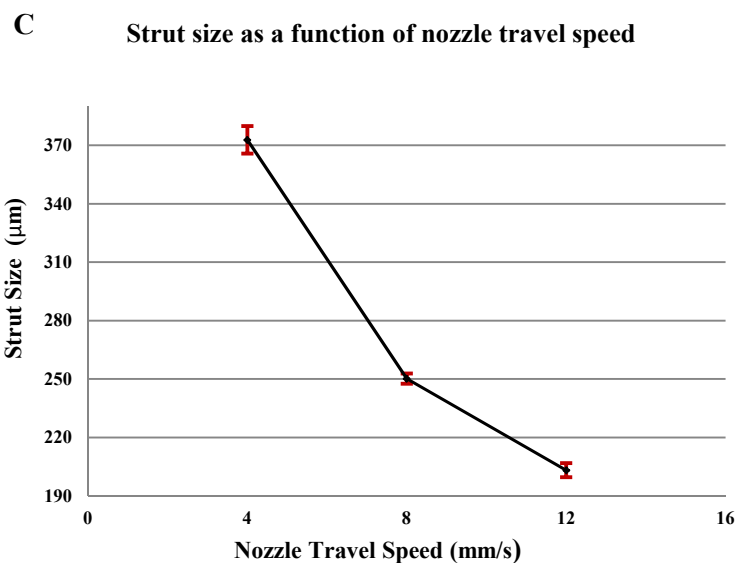


Figure 3. Effect of deposition parameters on strut formability. (A) Strut size as a function of alginate flow rate; (B) strut size as a function of dispensing tip size; (C) strut size as a function of nozzle travel speed.

3.2. Cross-linking parameter investigation

Effect of cross-linking parameters (alginate concentration, CaCl_2 concentration, and CaCl_2 to alginate flow rate ratio) on strut formability was examined. The fabricated struts were examined using Leica optical microscope with $10\times$ magnifications. The results were shown in Figure 4.

The result of alginate concentration examination was shown in Figure 4A. Strut sizes were measured as $288.00 \pm 4.06 \mu\text{m}$, $250.20 \pm 2.59 \mu\text{m}$ and $250.20 \pm 2.59 \mu\text{m}$. Examination of CaCl_2 concentration and CaCl_2 to alginate flow rate ratio was shown in Figure 4B. The struts measurements were shown in Table 1.

The strut formability of $250 \mu\text{m}$ was used as the criterion for cross-linking sufficiency examination. Considering the fluctuation caused by pneumatic pressure and other environmental disturbance, the acceptable struts sizes range of cross-linking sufficiency was set to $240 - 260 \mu\text{m}$. Sufficient cross-linking can be reached by alginate concentration: 4.5% and 6.0%, when other parameters were kept as base. Sufficient cross-linking range of CaCl_2 concentration and CaCl_2 to alginate flow rate ratio were shown in Figure 5 when other parameters were kept as base.

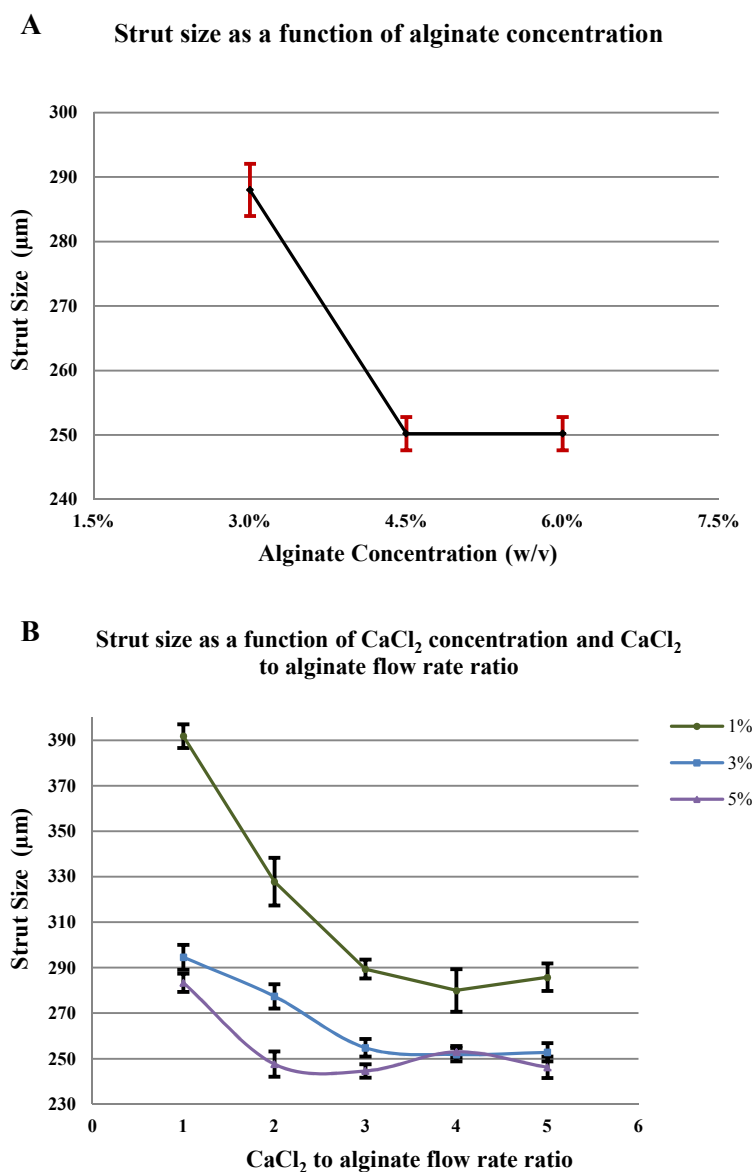


Figure 4. Effect of cross-linking parameters on strut formability. (A) Strut size as a function of alginate concentration; (B) strut size as a function of CaCl₂ concentration and CaCl₂ to alginate flow rate ratio.

Strut size (μm) as a function of CaCl ₂ concentration and CaCl ₂ to alginate flow rate ratio										
CaCl ₂ Concentration (w/v)	CaCl ₂ to alginate flow rate ratio									
	1:1		2:1		3:1		4:1		5:1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1%	391.8	5.22	327.8	10.47	289.4	4.16	280.0	9.41	285.8	6.06
3%	294.6	5.50	277.4	5.32	254.8	3.83	251.8	2.95	252.8	4.02
5%	283.4	4.04	247.6	5.50	244.6	2.97	253.0	2.55	246.2	4.66

Table 2. Strut size as a function of CaCl₂ concentration and CaCl₂ to alginate flow rate ratio

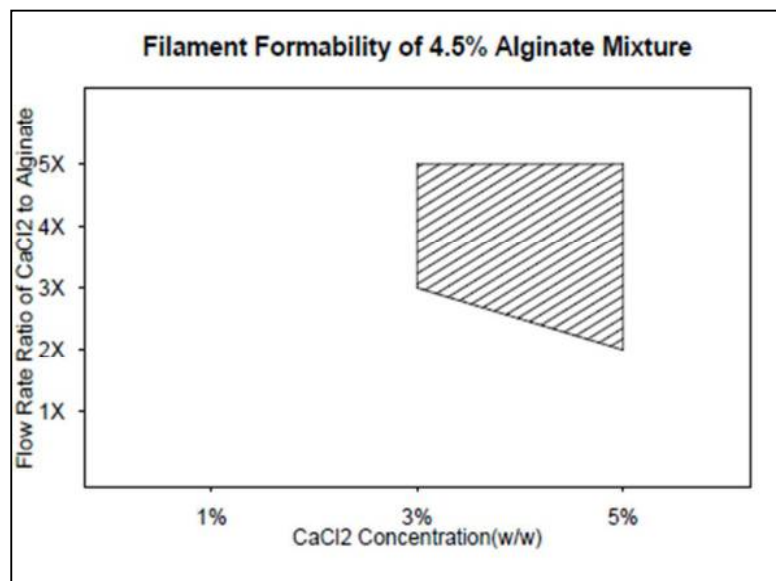


Figure 5. Sufficient cross-linking range (shadowed area) of CaCl_2 concentration and CaCl_2 to alginate flow rate ratio when other parameters were kept as base

3.3. Structural stability and integrity investigation

The material composition and cross-linking parameters achieved the strut formability were carried on for structural stability and integrity investigation. 3D scaffolds were fabricated and examined in regarding to structural stability. Results were shown in Figure 6. As observed, 3D scaffolds fabricated by using mixing ratio of 3:1 (alginate A: alginate B) and alginate A only held their structures during mechanical disturbance. Structures fabricated by other examined alginate mixing ratio fell apart by mechanical disturbance. All 3D scaffolds when examining cross-linking parameters held their structures during mechanical disturbance examination.

The constructs that survived the stability test were continued for examination of structural integrity in cell culture condition. Results were shown in Figure 7. As observed, 3D scaffolds fabricated by alginate A only loosed their integrity at day 3. Structures fabricated by alginate mixing ratio of 3:1 with all examined cross-linking parameters preserved their structures for 14 days in cell culture condition.

A quantitative study was carried out to confirm the structural integrity results. Strut diameters were measured on 3D scaffolds those passed the qualitative study. Results were shown in Table 3 and the trends were shown in Figure 8.

Examination of Material Composition on Stability against Mechanical Disturbance

Before Disturbance

After Disturbance

Examination of Cross-linking Parameters on Stability against Mechanical Disturbance

Before Disturbance

After Disturbance

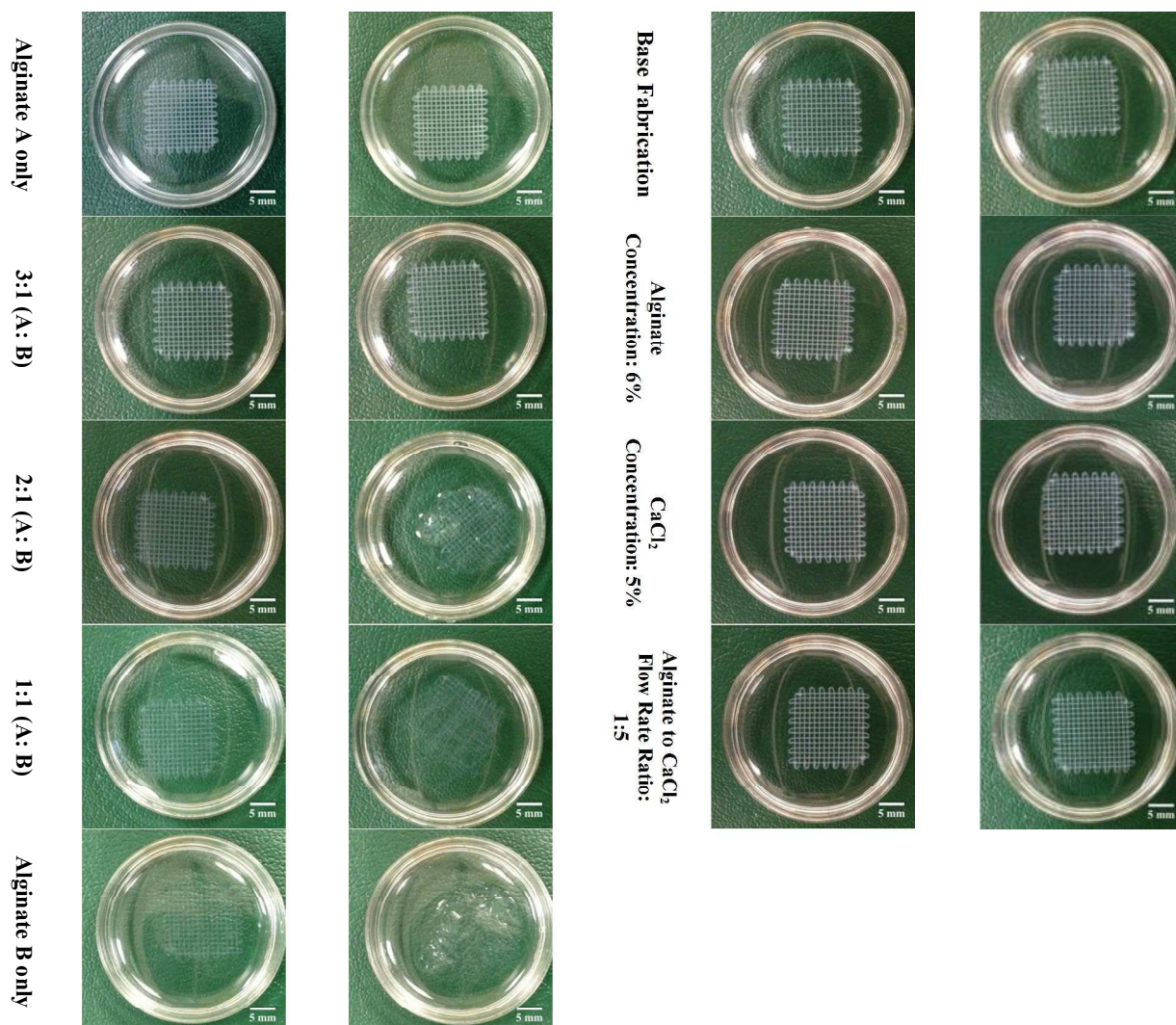
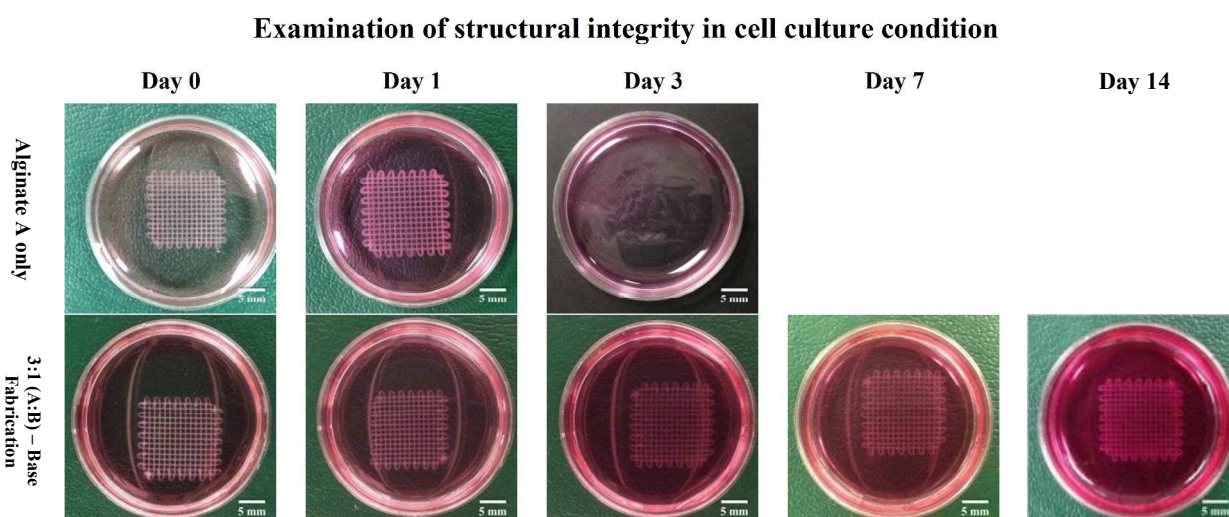


Figure 6. Examination of 3D laden structural stability against mechanical disturbance



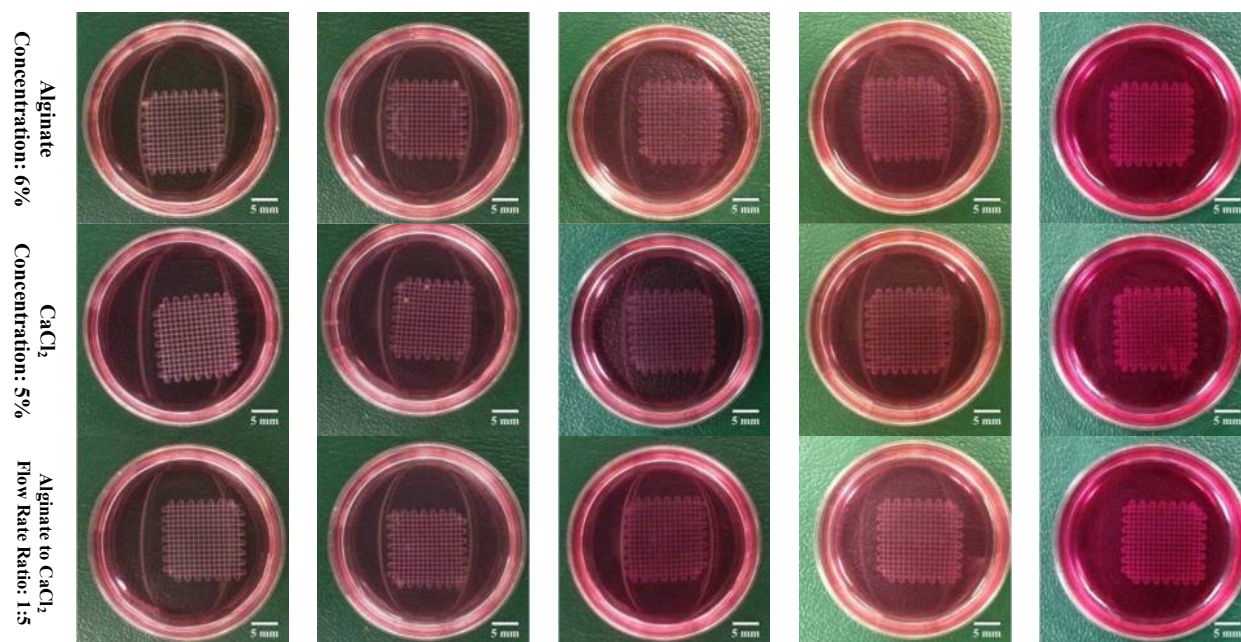


Figure 7. Examination of 3D laden structural integrity in cell culture condition

Strut size as a function of days incubated in cell culture condition						
	Strut Size (μm)					
	Day 0	Day 1	Day 3	Day 5	Day 7	Day 14
Base Fabrication Parameters	250.78 ± 3.46	221.11 ± 5.49	220.11 ± 5.53	219.78 ± 4.27	213.56 ± 4.19	216.22 ± 3.03
CaCl ₂ Concentration: 5%	250.89 ± 3.92	217.78 ± 3.60	217.22 ± 3.77	218.00 ± 3.94	213.44 ± 4.16	213.89 ± 2.57
Alginate Concentration: 6%	251.11 ± 4.20	232.11 ± 3.37	224.00 ± 2.78	224.11 ± 4.57	218.00 ± 4.77	216.78 ± 5.78
CaCl ₂ to Alginate Flow Rate Ratio: 5:1	253.22 ± 6.26	223.11 ± 5.88	219.11 ± 4.99	218.89 ± 7.66	213.89 ± 3.86	211.44 ± 3.36

Table 3. Strut size as a function of days incubated in cell culture condition

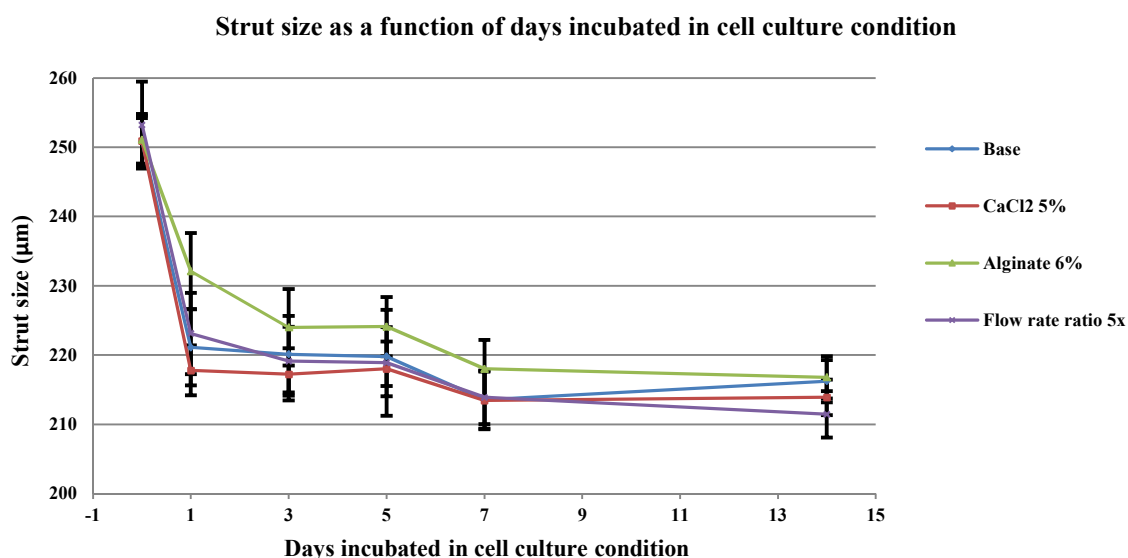


Figure 8. Strut size as a function of days incubated in cell culture condition

4. Discussion

Struts are the basic elements of 3D scaffold. The geometry and size of struts determine construct features as construct aspect ratio, pore size, nutrition/ waste exchanging rate, construct degradation rate, etc. These construct features are critical to the application in biomedicine. Thus, it was necessary to investigate the strut formability (the capability to fabricate a cylindrical strut in the same diameter as dispensing tip) of in situ printing. Strut fabrication was a process involved both alginate deposition and in situ cross-linking. Strut formability was investigated in terms of these two parametric groups: deposition related parameters and cross-linking related parameters.

Deposition related parameters were: alginate flow rate, dispensing tip diameter and nozzle travel speed. Sufficient cross-link was provided as 4.5% alginate concentration, 5% CaCl₂ suspension and 5:1 CaCl₂ to alginate flow rate ratio. Theoretically, strut formability can be achieved when alginate flow rate and nozzle travel speed were set according to dispensing tip diameter. The relationship of the three deposition parameters was developed as equation (1). Thus, strut formability in regarding to deposition parameters was investigated as the examination of equation (1) validity. Strut diameter was 250.20 ± 2.59 μm when deposition parameters satisfied equation (1). This result suggested that strut formability was achieved when equation (1) was satisfied. When equation (1) was not satisfied, strut diameter was significantly different from dispensing tip diameter. It was also interesting to see that the struts diameters were greater than 250 μm when altering the dispensing tip diameter to a greater value or a smaller value. This result suggested that cylindrical geometry can only be achieved as the equation (1) was satisfied. As dispensing tip diameter was altered, it was conjectured that a flattened cross-sectional geometry would be formed instead of cylindrical geometry. This flattened cross-sectional geometry would result in increased strut size measured by optic microscope.

Deposition parameters that satisfied equation (1) were utilized for cross-linking parameter investigation. Sufficient cross-linking was defined as sufficient numbers of physical cross-linking bonds that were able to mechanically hold the strut in cylindrical geometry. The sufficient numbers of cross-

linking bonds were determined by the total available cross-linking sites and cross-linker (calcium ions) availability. The cross-linking related parameters are: alginate suspension concentration, CaCl_2 suspension concentration and CaCl_2 to alginate flow rate ratio. As confirmed in deposition parameter investigation, strut formability can be achieved when cross-linked sufficiently. Thus, the strut formability was utilized as the criterion to determine the sufficient cross-linking range. During the examination of alginate concentration, strut formability was achieved when alginate concentration was 4.5% and 6%. This result suggested that an appropriate level of alginate concentration need to be reached for sufficient cross-link. Failed to reach this level would result in a low concentration alginate with insufficient available cross-linking sites. The availability of calcium ions was determined by CaCl_2 suspension concentration and CaCl_2 to alginate flow rate ratio. Hence, these two parameters were examined in pair combination. As shown in the results, strut formability can be achieved when the combination of the CaCl_2 concentration and CaCl_2 to alginate flow rate ratio reached to a level. This result suggested that calcium ion availability plays a role in strut formability. The cut off levels to achieve strut formability were utilized to determine sufficient cross-linking range for respective cross-linking parameters.

In order for *in situ* printing to be utilized as an efficient tool to fabricate 3D cell-laden construct for biomedical application, it is practically necessary that the fabricated 3D cell-laden constructs are able to sustain their structure against mechanical disturbances caused by various assays. In addition, biological reaction is a time depending process. Therefore, it is also necessary to have a 3D cell-laden constructs that can preserve their structure in cell culture environment for at least two weeks. Investigations of structural stability and integrity were performed to assess the capability to fabricate 3D scaffolds with those features.

Two types of alginates were utilized in this study: alginate A and alginate B. Alginate A had low molecular weight and alginate B had medium molecular weight. The two alginates were mixed and dissolved in 0.1M HEPES as alginate suspension. Mixing ratios were investigated as material composition in regarding to structural stability and integrity. The cross-linking parametrical values that provided sufficient cross-link were also investigated to study their effect on stability and integrity. It was found that the structural stability and integrity were only affected by the alginate mixing ratio. This result suggested that as cross-linked sufficiently, the cross-linking parameters did not affect the stability and integrity. 3D scaffolds fabricated by alginate A only and alginate mixing ratio of 3:1 (w/w) (alginate A: alginate B) sustained their structure during stability examination. This result suggested that constructs fabricated with more alginate A (low molecular weight) content were more stable mechanically. The constructs that survived stability test were carried on for integrity examination. It was found that construct fabricated by alginate A only was degraded at day 3 of incubation. Constructs those fabricated by alginate mixing ratio of 3:1 (w/w) (alginate A: alginate B) preserved their structure for at least 14 days. This result suggested that 3D constructs fabricated by more alginate B (medium molecular weight) content had lower degradation rate. A quantitative study was further performed to confirm this finding. As shown in Figure 8, significant degradation occurred at day 1. The strut sizes of 3D scaffolds fabricated by alginate mixing ratio of 3:1 (w/w) (alginate A: alginate B) were maintained above 200 μm for 14 days.

5. Conclusion

Alginate hydrogel 3D scaffolds can be successfully fabricated in an ambient condition. A range of fabrication process parameters were determined for desired strut size formation. It was found that the structural stability and integrity was only affected by alginate composition. Scaffolds with desired strut size, decent structural stability and structural integrity in cell culture condition for 14 days were fabricated.

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